DS202.3 – Data Programming with R
Coursework Assessment Report

Group C
36022 - Kaarthiraghav SHIVAGANESH
36484 - M.F.S. ZAINAB

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1 Problem Statement

The analytical question addressed in this research is: **Can basic patient demographics (age, gender) and clean clinical features (tumor type, stage) be used to predict a patient's primary treatment response?**

In modern oncology, predicting treatment response is a critical challenge that directly impacts patient longevity and quality of life. Cancer is an inherently heterogeneous disease; even within identical diagnostic classifications of tumor type and stage, individual biological responses to first-line treatments can vary significantly. This project utilizes a massive real-world dataset of 24,950 patients to explore whether routinely collected data like age and gender can serve as reliable indicators of medical outcome.

The global importance of this question lies in the transition toward precision medicine. If basic demographics can offer predictive insights, clinicians could rapidly identify patients who may not benefit from aggressive standard protocols, thereby avoiding ineffective treatments and unnecessary toxic side effects. This approach not only improves individual survival rates but also optimizes healthcare resource allocation by reducing costs associated with trial-and-error therapy.

2 Requirements & Design

The design of the analytical workflow prioritized data integrity and the removal of noise inherent in real-world clinical datasets, particularly regarding NLP-derived fields.

2.1 Data Cleaning and Manipulation

The data preparation phase was exhaustive, addressing significant quality issues before analysis commenced:

- **Syntactic Standardization**: Column names were converted to a consistent lowercase format without spaces using specific R functions to prevent syntax errors in the programming environment.
- **Handling Missing Values**: Multiple "Unknown" and "[Not Available]" string formats were identified and converted to true NA values. We implemented a strict 50% threshold rule, identifying and dropping any columns where missing data exceeded half of the observations to ensure the robustness of the final feature set.
- **Type Conversion**: The age variable was validated and converted to a numeric format, while clinical categorical features were factorized for proper statistical handling.

2.2 Notebook Structure and Tooling

Our R Notebook is organized into logical segments: Environment Setup, Data Ingestion, Cleaning/Transformation, and Multi-stage Exploratory Data Analysis (EDA). We utilized the tidyverse suite—specifically packages for manipulation and visualization—due to its cohesive syntax and efficiency in handling data frames of this scale.

3 Implementation & Results

3.1 Analytical Process

We executed a hierarchical analysis starting with Univariate distributions to understand the cohort's baseline characteristics, such as the peak age range being 50–80 years. This was followed by Bivariate analysis to investigate the relationship between gender and specific cancer types, and finally, Multivariate analysis to examine how age and tumor stage interact relative to the primary response.

3.2 Key Findings and Visualizations

The analysis revealed that demographic variables like age and gender are surprisingly weak predictors of treatment outcome when viewed in isolation.

The visualizations demonstrate that age distribution remains remarkably stable across both "Responded" and "Non-Responded" groups. This suggests that while clinical features like tumor type and simplified stage are essential for a predictive foundation, demographics alone are insufficient for high-accuracy patient care models.

4 Personal Reflection

Working with R libraries was highly effective for managing a dataset of nearly 25,000 records. The most challenging aspect was the "messy" nature of the NLP fields, which initially led to runtime errors until we standardized our NA handling. A key lesson learned is the absolute necessity of rigorous data profiling before attempting visualization; without the 50% threshold rule for missing values, initial plots were skewed and misleading.

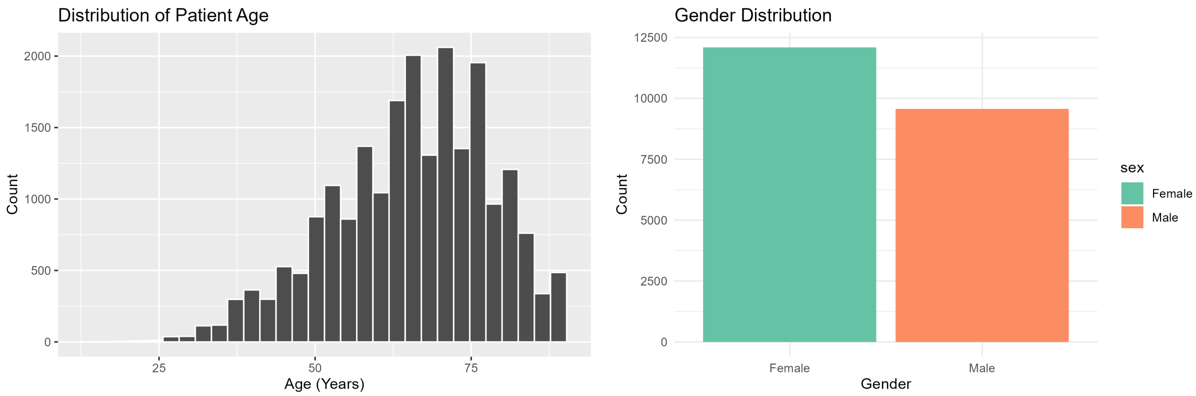

Figure 1: Distribution of Patient Age and Gender within the cohort.

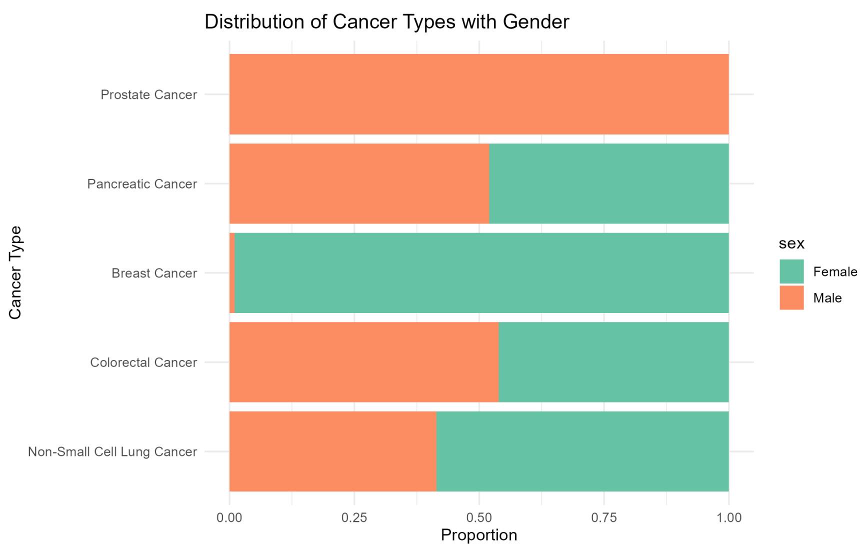

Figure 2: Distribution of Cancer Types across Genders.

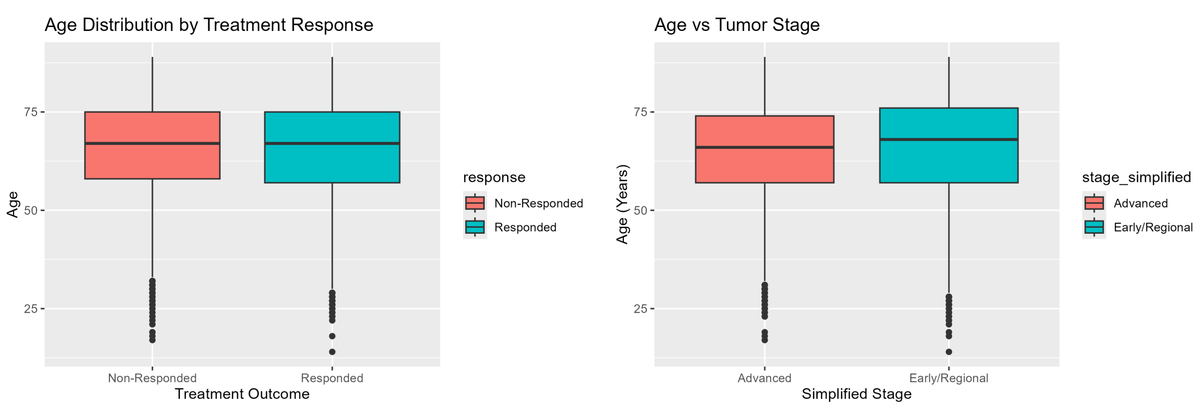

Figure 3: Multivariate Analysis: Age by Response and Age by Tumor Stage.

5 Future Work

This analysis could be significantly improved by integrating genomic and molecular features present in the sequencing data, such as mutation counts and tumor purity. To make this notebook more scalable, we would add automated unit tests to the cleaning pipeline to validate data types upon ingestion, ensuring the analysis remains sustainable as new patient records are added.

6 Conclusion

Exploratory data analysis reveals that basic demographics like age and gender are weak predictors of treatment outcome when used in isolation. However, clinical features like tumor type and simplified stage provide a necessary foundation for predictive modeling. A successful model must prioritize structured clinical and genomic features to achieve high accuracy in patient care.

7 Project Links

- **GitHub Repository**: https://github.com/Kaarthiraghav/DPR-Group-C
- **Video Demonstration (YouTube)**: https://youtu.be/bw4nb-yX$_MA$